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DA 401

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20 September 2026

The Home Run Derby Curse: Second-Half Performance, Swing Approach, and Methods Used to Study Them

Major League Baseball implemented the Home Run Derby as an event during the All-Star Break in 1985, bolstering the appeal of the “Midsummer Classic.” The first instance of the derby featured 10 players, 5 from each league, all of whom were considered some of the best players in baseball at that time. The event became one of the most anticipated across sports over the next handful of years, and shed more light on some of the best players to ever set foot on a baseball field. However, as the Derby became more established, concerns grew about potential injury risk and decline in performance among players who participated, or a Home Run Derby curse (O’Leary, 2016). As data and statistics became more publicly available, efforts to research this theory became more prevalent. This research mainly addressed talks of a potential decline in performance after participating; however, experts were also met with questions about injury risk (DeFroda et al., 2021).

This review is organized by theme. First, it summarizes what studies of second-half performance have found, and how regression to the mean has shaped those findings. The, it turns to what those studies leave out, mainly covering batted-ball results and hitter approach, and reviews research on batter swings. A skill review follows, which covers statistical techniques used in this area, and the review closes with the gaps in research that remain. Injury risk is

outside the scope of this review, though DeFroda et al. (2021) found no significant association between Derby participation and second-half injury.

Second-Half Performance: Is There a Decline?

Early quantitative work tended to show that Derby participants did worse after the break, but interpretation of that decline is where the debate begins. Jaiclin and McCollum (2010) examined 1,111 player seasons through 2009, with 192 of these seasons involving Derby participation. In doing so, they found that participants' OPS decreased by .043 from the first half to the second half. However, when they replaced those seasons with statistically comparable seasons in which no Derby was played, the decline was identical (.043), allowing them to conclude the curse was fiction. O'Leary (2016) reached a more mixed result when applying data from 1999 to 2013. Competitors performed significantly worse on four key analytics, yet Derby winners showed no significant difference before and after this event, with O'Leary framing the results through both regression to the mean and winner and loser effects. More recently, DeFroda et al. (2021) compared 114 Derby participants with 114 All-Star Game players from 2006 to 2019, finding no significant differences in batting average, on-base percentage, slugging percentage, strikeouts, or home runs. The only difference was participants producing a lower season-long WAR (4.69 vs 5.33) that the authors said unmeasured confounders could explain. Sobol et al. (2024) found a similar pattern in the NBA Slam Dunk Contest, with no decline in production among 88 participants compared with 88 All-Star controls. While the Dunk Contest poses fewer potential changes in athlete mechanics than a Home Run Derby, it is still worth considering similar circumstances in other major sports. Taken together, a decline appears in the raw numbers, but as comparison groups improve, evidence of the Derby causing the decline grows weaker.

Regression to the Mean and Selection Bias

The central methodological issue is that Derby participants are not a random sample. Players are typically invited because of strong first-half performance, and Barnett et al. (2005) explain that when a group is chosen because of extreme values, later measurements tend to move back toward the average even if nothing about the players has changed. A second-half decline is therefore what would be expected regardless of the Derby. Schall and Smith (2000) treat this as an empirical question for MLB hitters, asking directly whether players regress toward the mean, which is a reminder that the effect has to be tested for rather than assumed.

Because of this, how researchers build a comparison group matters as much as the results themselves. Jaiclin and McCollum (2010) built comparable non-Derby seasons, while DeFroda et al. (2021) selected All-Star Game players with the highest first-half home run totals, excluding Derby participants, and argued that “similarity score” controls introduce significant selection bias. Both approaches try to give the control group the same extreme first-half selection so that regression to the mean affects both groups equally. The studies still differ in how much weight they deserve. Jaiclin and McCollum’s data end in 2009, before the timer-based format, and their approach depends on how similarity is defined. O’Leary’s winner group is small, with only one winner per year. DeFroda et al. is the most careful about control selection but remains observational and cannot establish causation. Overall, the stronger the control design, the weaker the case for a curse.

What Prior Studies Leave Out: Batted-Ball Results and Swing Approach

Nearly all of the studies above judge the Derby’s effect through outcome statistics such as OPS, batting average, and home runs. Those numbers depend on many things beyond the hitter’s swing, including pitching, defense, and luck, over a second half of only about 65 to 70 games, so

a real change in how a hitter swings can be hidden or exaggerated. Batted-ball data such as exit velocity, launch angle, and the direction the ball is hit describe what the hitter is doing more directly, yet none of the Derby studies reviewed here analyze them, and the most recent Derby-specific study found in this search was published in 2021.

There is a plausible mechanical reason to look there. Success in the Derby typically relies on pulling the ball in the air, which requires making contact earlier in the ball's flight, farther out in front of the body. Welch et al. (1995) describe hitting as a sequence of coordinated phases, and Nakashima et al. (2025) found that the acceptable timing window at impact averaged only 9.36 ms and depended on vertical swing angle, horizontal swing angle, and bat inclination, meaning small changes in swing path leave little room for error. Giordano et al. (2026) analyzed 2,779 batted balls from 239 collegiate hitters and found that contact depth was closely tied to attack angles, with vertical and horizontal attack angles together explaining 80% of its variance. Taken together, these findings suggest that a lasting change in contact point could show up in batted-ball direction and launch angle even when OPS holds steady. These studies used laboratory or collegiate samples and none tested the Derby, so whether a Derby-style approach carries into the season remains untested.

The event is also changing. Major League Baseball (2026) announced that the 2026 Derby would drop the timer used since 2015 and return to a swing-based format with 20 swings in the first round and 15 in each later round, with players who homer on their final swing continuing until they miss. Because format determines how many high-intent swings a participant takes, studies that pool Derbies across decades combine different formats, a limitation the studies reviewed here do not appear to address.

Skill Review: Methods Used in This Area

Research on the Derby curse relies on a small set of tools. The simplest is the comparison of a statistic such as OPS between the two halves of a season, tested for significance, as in Jaiclin and McCollum (2010) and O’Leary (2016). Confidence intervals add a range of plausible values for a change, so a small decline can be judged against its uncertainty and not only its p-value. A step up is the matched-control design with multivariable regression, which DeFroda et al. (2021) used to compare hitting performance between groups and which Sobol et al. (2024) applied in basketball.

Clustering is common in baseball analytics for grouping players by profile. Camp et al. (2020) used k-means clustering with the elbow method on Statcast data to define pitch subtypes from movement, velocity, and spin. Oda and Hirotsu (2026) grouped 72 Nippon Professional Baseball position players using 115 hitting performance indices, including batted-ball measures. Because so many of the indices were correlated, they first applied PCA, retaining about 70% of the cumulative variance, and then clustered the component scores. Koseler and Stephan (2017), in their systematic review of machine learning in baseball, organized the literature around supervised prediction problems (regression and classification), suggesting that grouping hitters by profile is a less explored use.

PCA is certainly suitable for this kind of data. Abdi and Williams (2010) describe PCA as a technique that converts a set of correlated variables into a smaller set of uncorrelated components ranked by how much variance they explain, with loadings that show which original variables drive each component. Batted-ball measures including exit velocity, launch angle, pull rate, and fly-ball rate are related to one another, and loadings can show which carry the most information and which are redundant. Jolliffe and Cadima (2016) stress the practical decisions involved, including how many components to keep, whether to standardize variables measured

on different scales, and how to interpret components, and caution that components capture variance and not necessarily meaning. PCA also does not correct for regression to the mean, which is important to consider due to any before-and-after comparison needing the control group logic described above.

Broader Relevance

The question matters beyond fan folklore. The answer bears on how players and teams view Derby participation, and batted-ball data could show whether any change in hitting reflects a real change in approach or only noise in outcome statistics. There has been research to disprove a statistically significant performance decline across the league. However, teams, players and fans deserve a more intense study of what Home Run Derby participation may imply for a specific player, with the millions of dollars being invested into players' success. The same combination of clustering, PCA, and controlled before-and-after comparison could be reused to study how other changes in approach affect hitters.

Conclusions

The research so far points in one direction. Second-half declines appear in raw statistics (Jaiclin & McCollum, 2010; O'Leary, 2016), but studies that account for regression to the mean and use better comparison groups find little evidence that the Derby causes them (DeFroda et al., 2021). Several questions remain open. Prior work relies on outcome statistics instead of batted-ball data, treats each statistic separately, pools Derbies across different formats, and offers little recent Derby-specific research. One next step is to use PCA and clustering to summarize hitters' batted-ball profiles before and after the Derby, then test the changes with regression, confidence intervals, and t-tests against a control group selected to account for regression to the

mean, paying attention to whether changes in pull rate and air-ball rate fit the mechanical explanation described above.

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